

UNIT 6: DIVERSITY OF LIFE FORMS

6.2 – THE PROKARYOTES

Prokaryote Outline

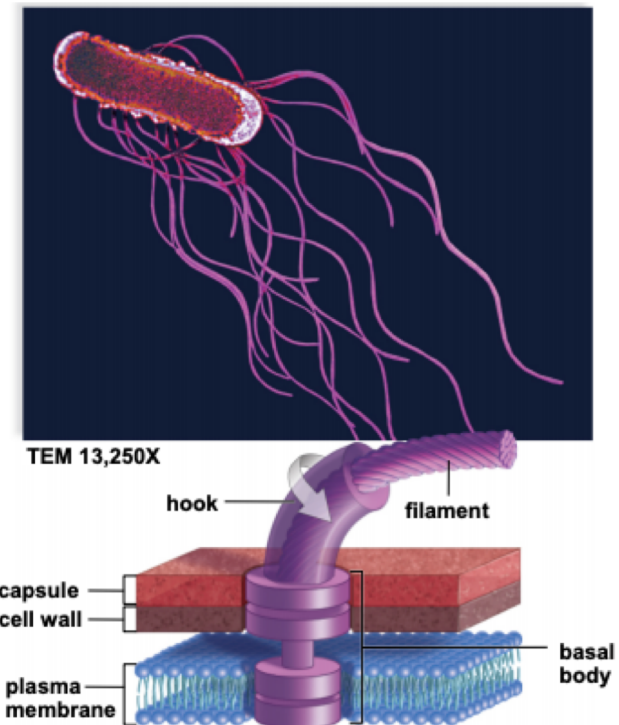
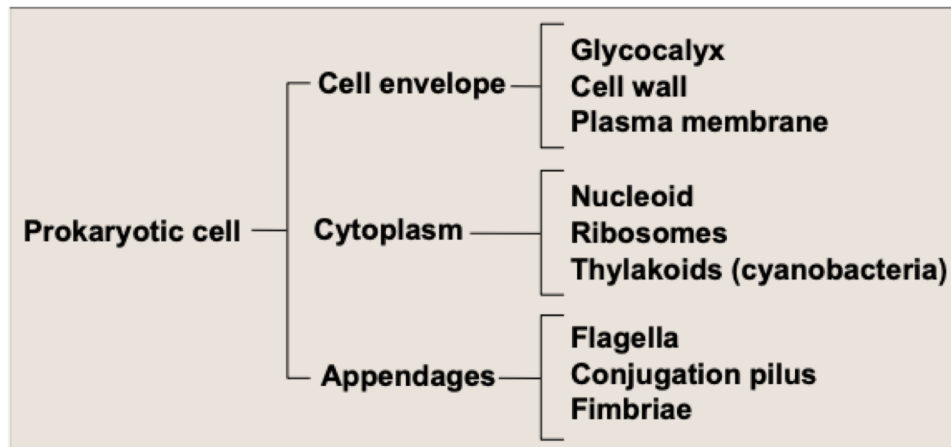
- Structure
- Reproduction
- Nutrition
- Classification
 - Bacteria
 - Archaea

Background

- Include bacteria and archaea, which are fully functioning cells
 - ▣ A single spoonful of earth can contain >1000 prokaryotes
 - ▣ Range in size from 1-10 μm in length and 0.7-1.5 μm in width
- Originally believed to be spontaneously generating organisms
 - ▣ Disproved in 1850's by Louis Pasteur

Structure

- ❑ Lack a membrane-bounded nucleus (DNA in nucleoid region)
- ❑ Outer cell wall containing peptidoglycan
- ❑ Some move by means of flagella
- ❑ Lack membranous organelles
- ❑ May have accessory ring of DNA (plasmid)



Reproduction

□ Asexual

- Prokaryotes reproduce asexually by means of binary fission

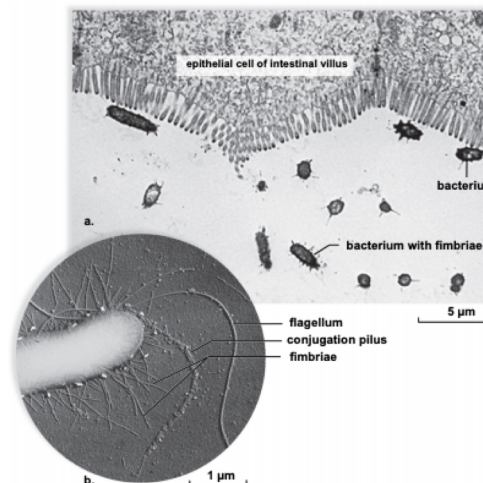
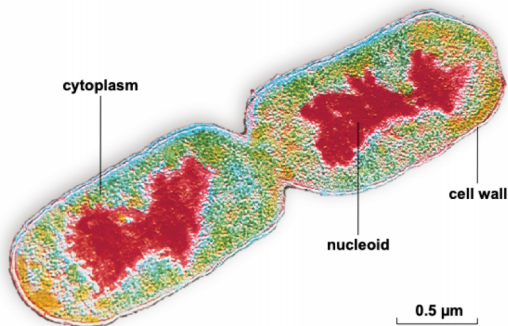
□ Methods of genetic recombination

■ Conjugation

- Sex pilus forms between two cells
- Donor cell passes DNA to recipient cell through pilus

■ Transformation

■ Transduction



Reproduction

□ Transformation

- Occurs when bacterium picks up free pieces of DNA from other prokaryotes
- Becomes incorporated into genome

□ Transduction

- Occurs when bacteriophages carry portions of bacterial DNA from one cell to another
- Serve as vectors

Nutrition

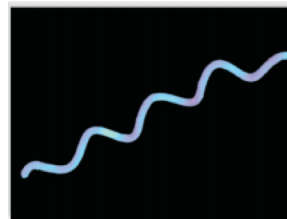
- Oxygen requirements:
 - Obligate aerobes – unable to grow in the absence of free oxygen
 - Obligate anaerobes – unable to grow in the presence of free oxygen
 - Facultative anaerobes – able to grow in either the presence or absence of free oxygen

Nutrition

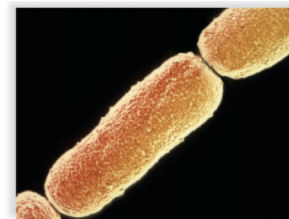
- Photoautotrophs
 - ▣ Use solar energy to reduce carbon dioxide to organic compounds
 - ▣ Photosynthetic
- Chemoautotrophs
 - ▣ Oxidize inorganic compounds to obtain the necessary energy
 - ▣ Use it to reduce CO₂ to an organic compound
 - ▣ Chemosynthetic
- Most prokaryotes are chemoheterotrophs that take in organic nutrients
 - ▣ Aerobic saprotrophs decompose most large organic molecules to smaller molecules
 - ▣ Essential components of healthy ecosystem
- May be free-living or symbiotic
 - ▣ Nitrogen fixation
 - ▣ Commensalism
 - ▣ Parasites

Domain Bacteria

- Bacteria are the more common type of prokaryote.
 - ▣ Over 2,000 different bacteria have been named.
 - ▣ Most bacterial cells are protected by a cell wall
 - Contains peptidoglycan
- Bacteria are commonly diagnosed using the Gram stain procedure
 - ▣ When washed after staining:
 - Gram-positive bacteria retain dye and appear purple
 - Gram-negative bacteria do not retain dye and appear pink
- Bacteria can be further classified in terms of their three basic shapes
 - ▣ Spiral (spirilli),
 - ▣ Rod (bacilli), and
 - ▣ Round (cocci)



a. Spirillum:
Spirillum volutans SEM 3,520X



b. Bacilli:
Bacillus anthracis SEM 35,000X



c. Cocci:
Streptococcus thermophilus SEM 6,250X

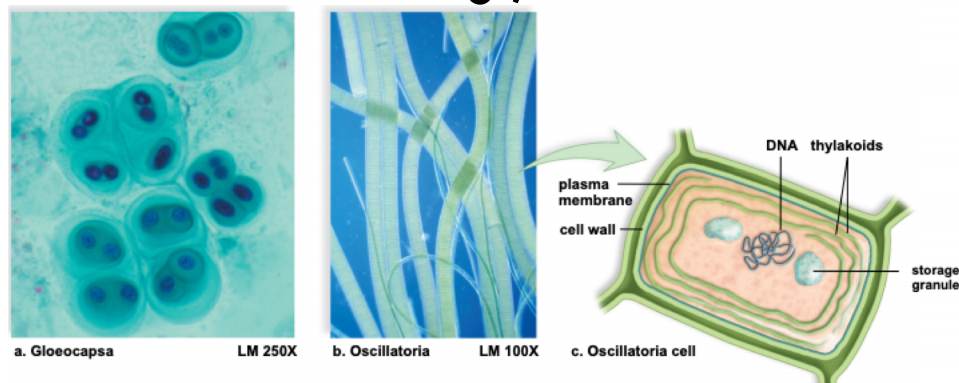
Domain Bacteria

Bacterial Diseases in Humans

<i>Category</i>	<i>Disease</i>
Sexually transmitted	Syphilis, gonorrhea, chlamydia diseases
Respiratory diseases	Strep throat, scarlet fever, tuberculosis, pneumonia, Legionnaires disease, whooping cough, inhalation anthrax
Skin diseases	Erysipelas, boils, carbuncles, impetigo, acne, infections of surgical or accidental wounds and burns, leprosy (Hansen disease)
Digestive tract diseases	Gastroenteritis, food poisoning, dysentery, cholera, peptic ulcers, dental caries
Nervous system diseases	Botulism, tetanus, leprosy, spinal meningitis
Systemic diseases	Plague, typhoid fever, diphtheria
Other diseases	Tularemia, Lyme disease

Domain Bacteria

- Cyanobacteria - Formerly called the Blue-Green algae (Cyanophyta)
 - ▣ Cyanobacteria are Gram-negative bacteria that photosynthesize
 - ▣ Believed to be responsible for introducing oxygen into the primitive atmosphere
 - ▣ Lack visible means of locomotion
 - ▣ Can live in extreme environments
 - ▣ When commensals with fungi, form lichens



Domain Archaea

- Archaea were earlier considered bacteria
 - ▣ Discovered that the base sequence of their rRNA differs from Bacteria
 - ▣ Other differences:
 - Archaea do not have peptidoglycan in their cell walls like the Bacteria
 - Archaea biochemical more like Eukarya than Bacteria
 - Archaea now thought to be more closely related to Eukarya than to Bacteria
- Most are chemoautotrophs
- Some mutualistic, Some commensalistic
- None known to be parasitic
- None are photosynthetic

Domain Archaea

- Many live in harsh conditions:
 - Anaerobic marshes
 - Methanogens
 - Produce methane from hydrogen gas and carbon dioxide
 - Salty lakes
 - Halophiles
 - Require high salt concentrations for growth, and
 - Hot sulfur springs & Thermal vents
 - Thermoacidophiles
 - Reduce sulfides and survive best at temperatures above 80°C and low pH
 - Plasma membranes contain unusual lipids convey tolerance of high temperatures

